During my first day of an introductory computer programming class, my teacher had mentioned a person who had heavily contributed to the world of computer science and is someone who has shown me the possibilities with what I am intending to learn. Maragaret Hamilton, a renowned software engineer that had heavily contributed to on-board flight software during NASA's Apollo program.

During the Apollo 11 moon mission, Hamilton chose to challenge the initial engineering and programming behind the spacecraft's software and architecture because she understood how high of a risk it was. Hamilton double checked security protocols and also discovered and implemented a protocol of priority scheduling. She was able to anticipate any human or systematic error that could get in the way of messing with the computer's system. Without her efforts to think ahead, the possibility of the computer in the spacecraft becoming overwhelmed was a very high possibility.

During the renowned Apollo 11 mission, her efforts were proved to have been so beneficial. Right before the renowned Neil Armstrong and Buzz Aldrin had officially landed on the moon, the radar system (already incorrectly configured) had started to malfunction and infiltrate the computers system with thousands of unnecessary calculations almost every second. This normally would have completely overloaded the computer's system and either caused it to freeze or crash and would have caused a very unsuccessful landing. Thanks to Hamilton's ability to plan ahead and correctly architectured system, the computer did not crash. Instead, the Apollo Guidance Computer was able to correctly recognize the error and displayed a 1202 error which gave the crew enough time to analyze the problem and divert the spacecraft's energy away from

the background tasks and to the landing and navigation systems. Her instincts and wisdom successfully helped not only the mission but also the world of programming.

After my teacher had informed me of such an incredibly integral part of the beginning of computer programming, I was immediately inspired, especially as a woman, to pursue programming and gain a greater understanding of how much application to the real world the smallest details can have.